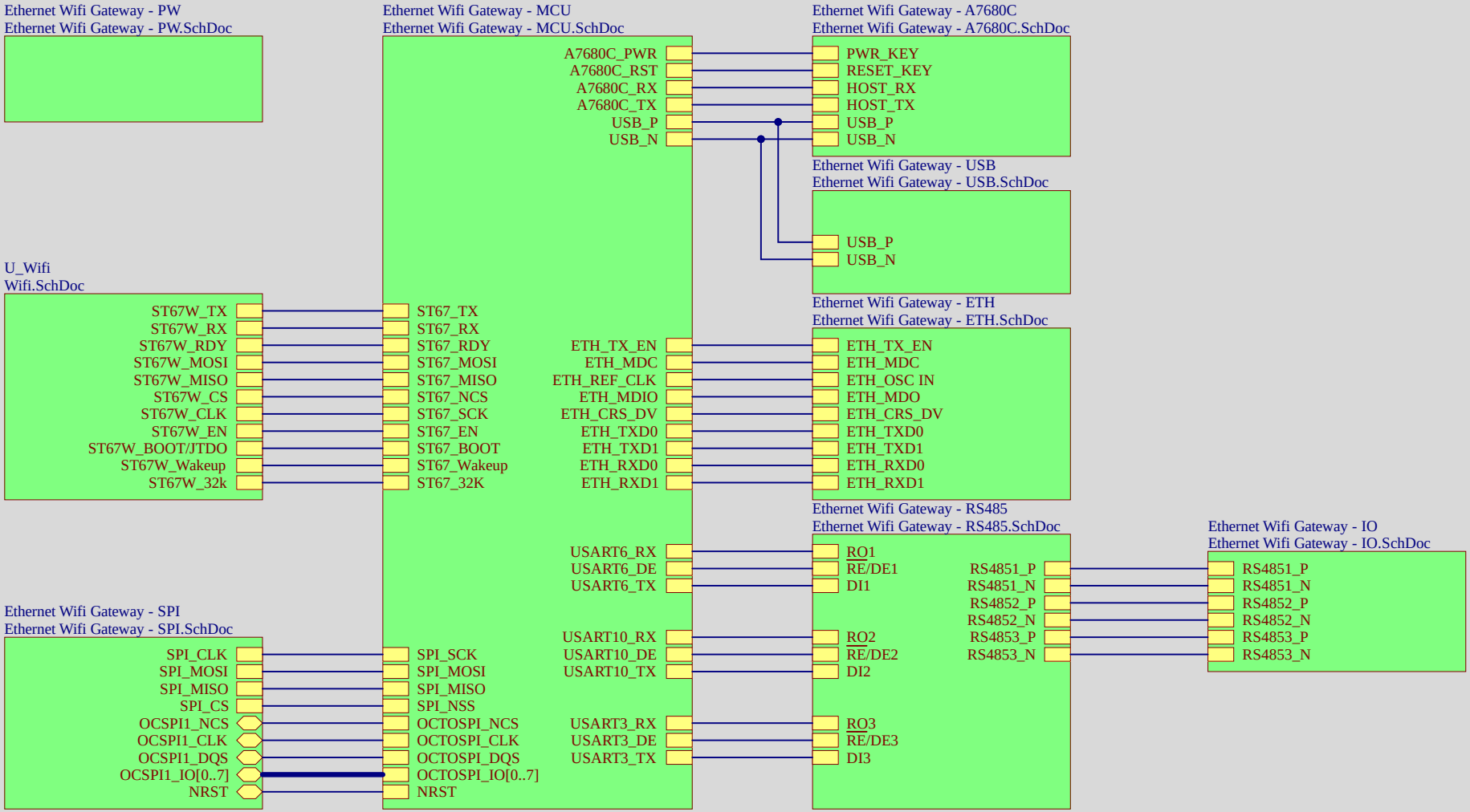
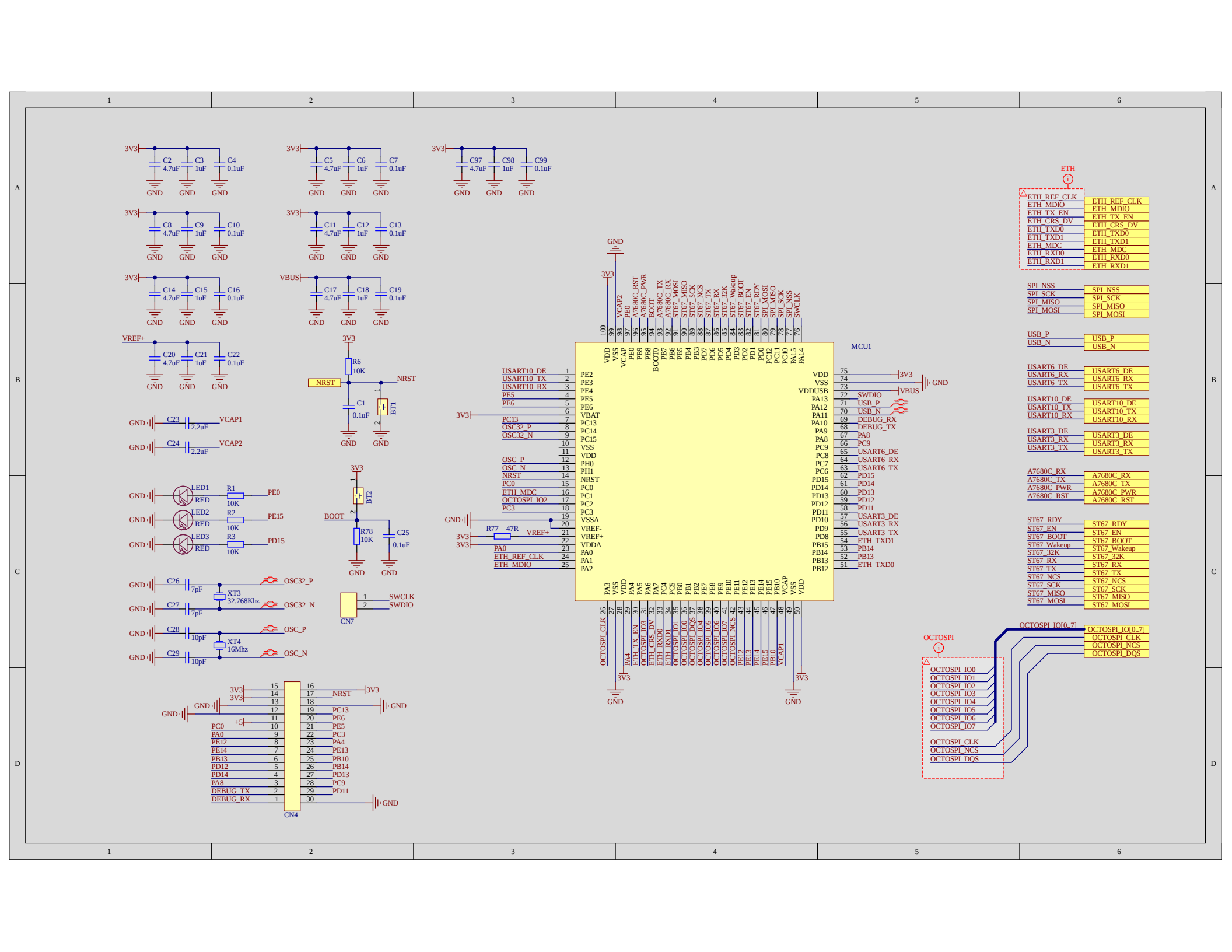
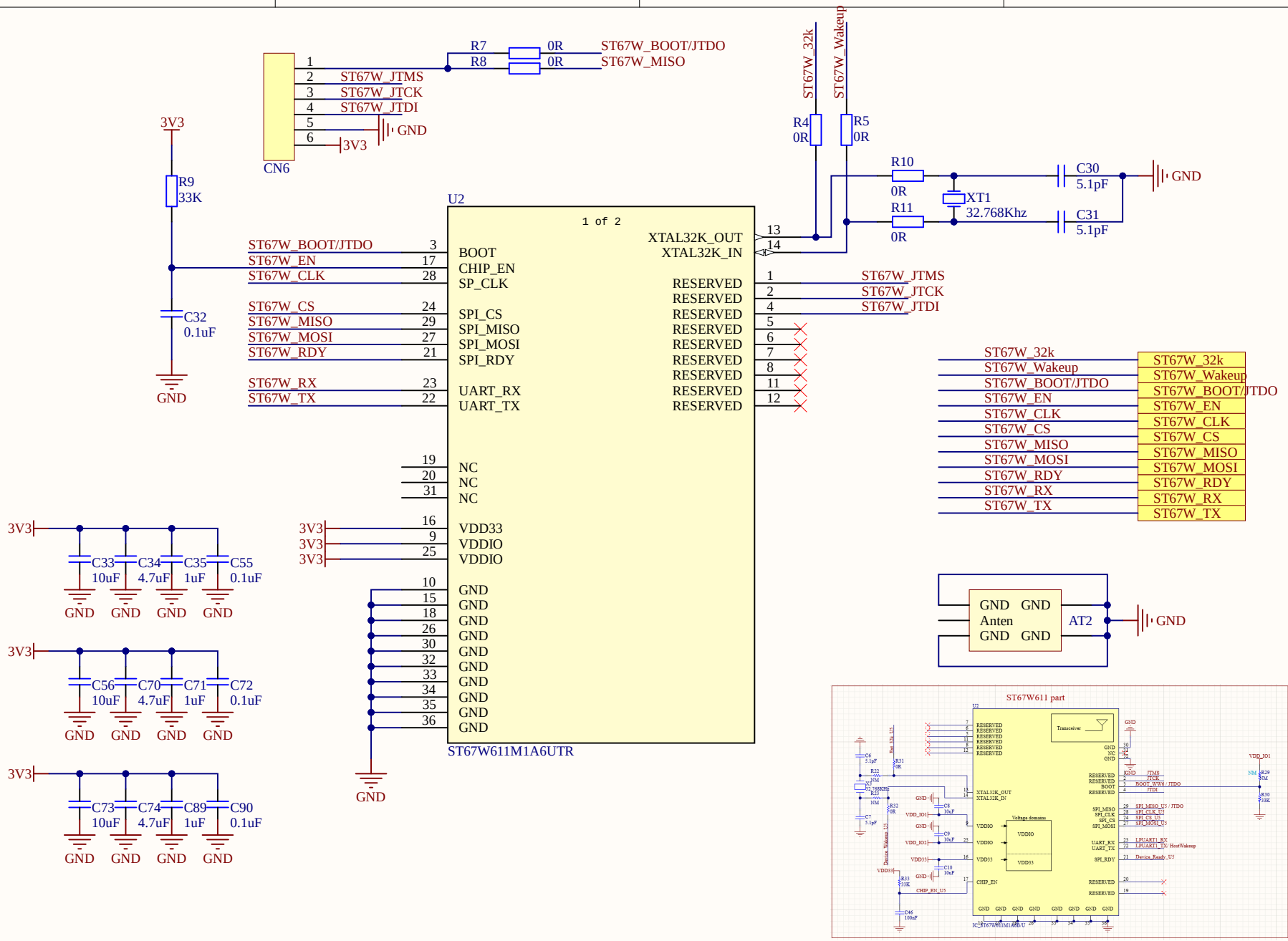
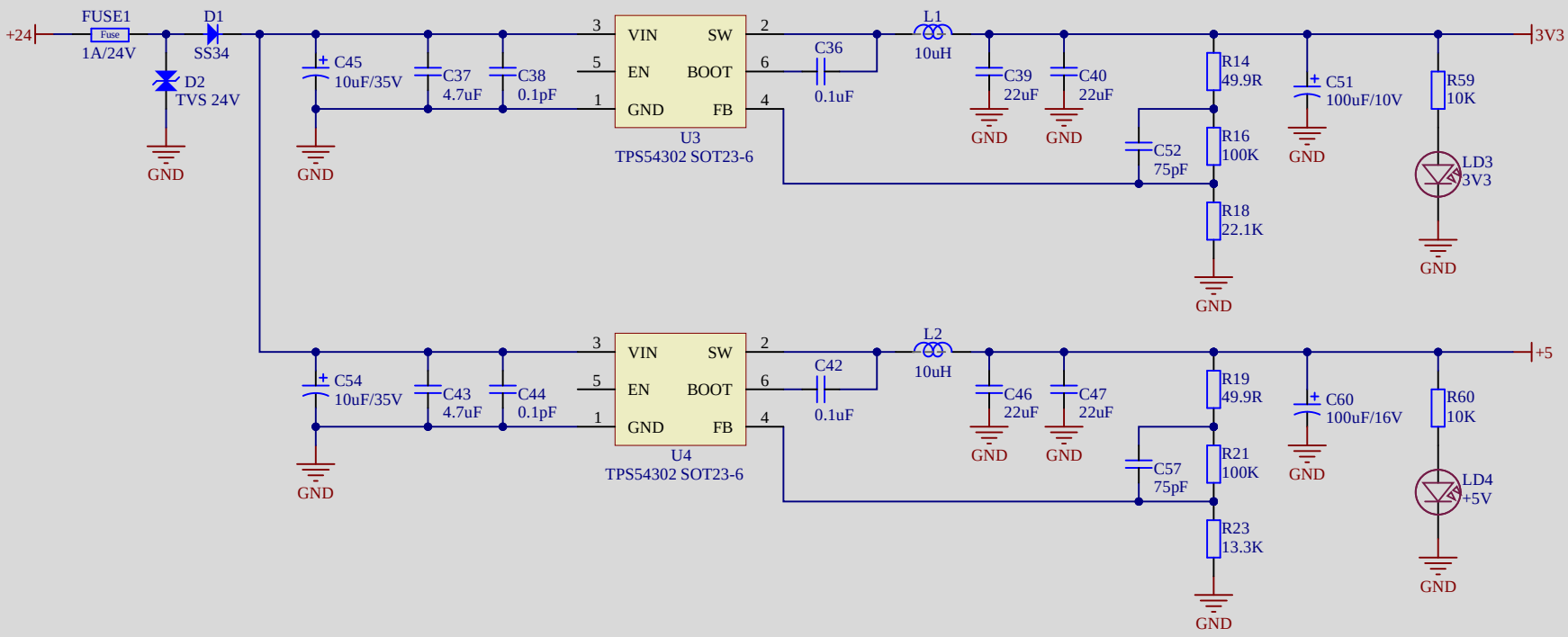
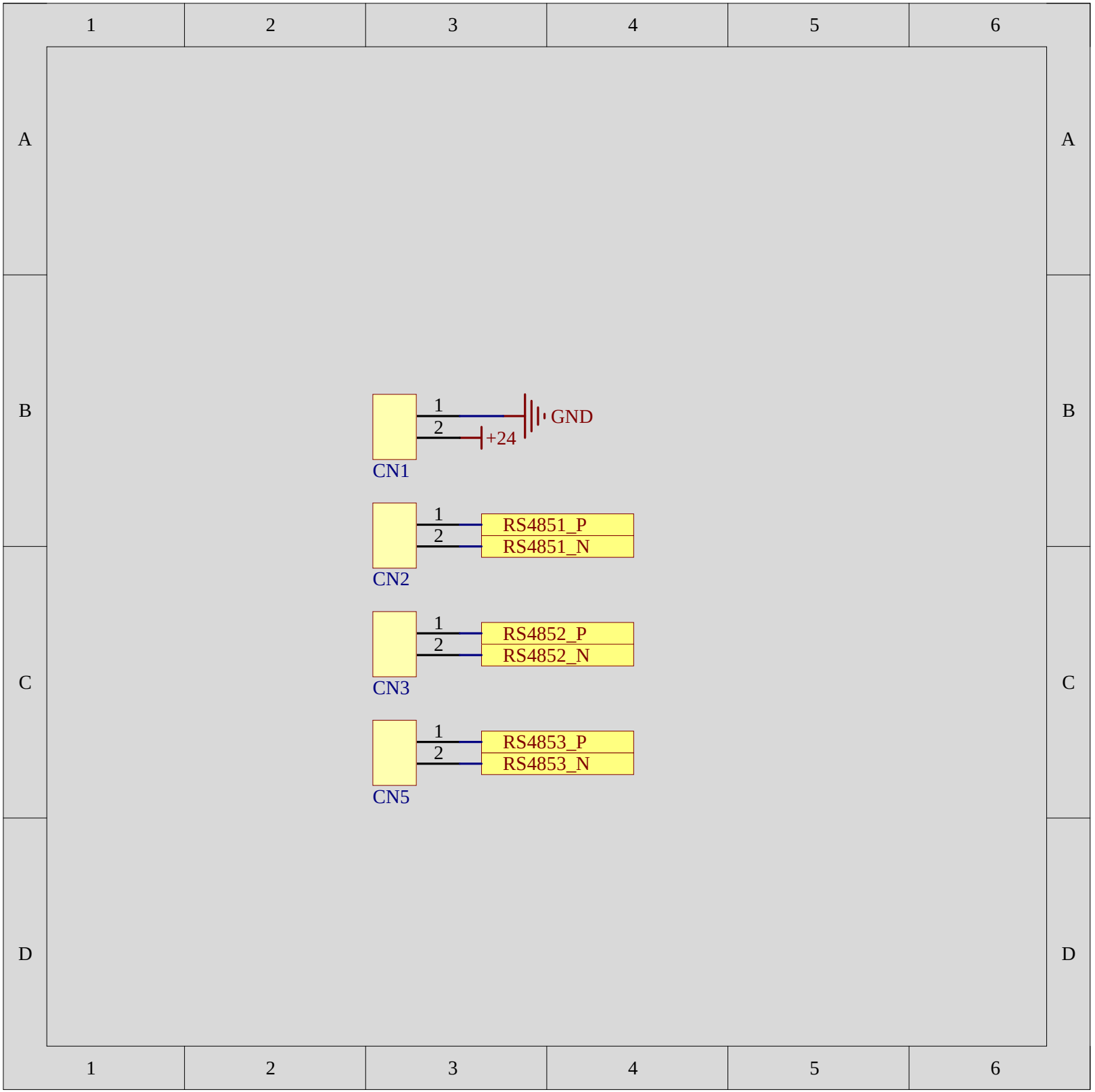

Ethernet Wifi Gateway - PW
Ethernet Wifi Gateway - PW.SchDoc
Ethernet Wifi Gateway - MCU
Ethernet Wifi Gateway - MCU.SchDoc
A7680C_PWR
A7680C_RST
A7680C_RX
A7680C_TX
USB_P
USB_N
Ethernet Wifi Gateway - A7680C
Ethernet Wifi Gateway - A7680C.SchDoc
PWR_KEY
RESET_KEY
HOST_RX
HOST_TX
USB_P
USB_N
Ethernet Wifi Gateway - USB
Ethernet Wifi Gateway - USB.SchDoc
USB_P
USB_N
U_Wifi
Wifi.SchDoc
ST67W_TX
ST67W_RX
ST67W_RDY
ST67W_MOSI
ST67W_MISO
ST67W_CS
ST67W_CLK
ST67W_EN
ST67W_BOOT/JTDO
ST67W_Wakeup
ST67W_32k
ST67_TX
ST67_RX
ST67_RDY
ST67_MOSI
ST67_MISO
ST67_NCS
ST67_SCK
ST67_EN
ST67_BOOT
ST67_Wakeup
ST67_32K
Ethernet Wifi Gateway - ETH
Ethernet Wifi Gateway - ETH.SchDoc
ETH_TX_EN
ETH_MDC
ETH_REF_CLK
ETH_MDIO
ETH_CRS_DV
ETH_TXD0
ETH_TXD1
ETH_RXD0
ETH_RXD1
ETH_OSC IN
Ethernet Wifi Gateway - RS485
Ethernet Wifi Gateway - RS485.SchDoc
USART6_RX
USART6_DE
USART6_TX
USART10_RX
USART10_DE
USART10_TX
USART3_RX
USART3_DE
USART3_TX
RO1
RE/DE1
DI1
RO2
RE/DE2
DI2
RO3
RE/DE3
DI3
RS4851_P
RS4851_N
RS4852_P
RS4852_N
RS4853_P
RS4853_N
Ethernet Wifi Gateway - IO
Ethernet Wifi Gateway - IO.SchDoc
Ethernet Wifi Gateway - SPI
Ethernet Wifi Gateway - SPI.SchDoc
SPI_CLK
SPI_MOSI
SPI_MISO
SPI_CS
OCSPI1_NCS
OCSPI1_CLK
OCSPI1_DQS
OCSPI1_IO[0..7]
NRST
SPI_SCK
SPI_MOSI
SPI_MISO
SPI_NSS
OCTOSPI_NCS
OCTOSPI_CLK
OCTOSPI_DQS
OCTOSPI_IO[0..7]
NRST
Title
Size
A
Number
Revision
Date: 7/06/2026
Sheet of
File: D:\OneDrive - MB376106-Cty CP Havicom Viet Nam\Havicom\..\Ethernet Wifi Gateway.SchDoc
Drawn By:

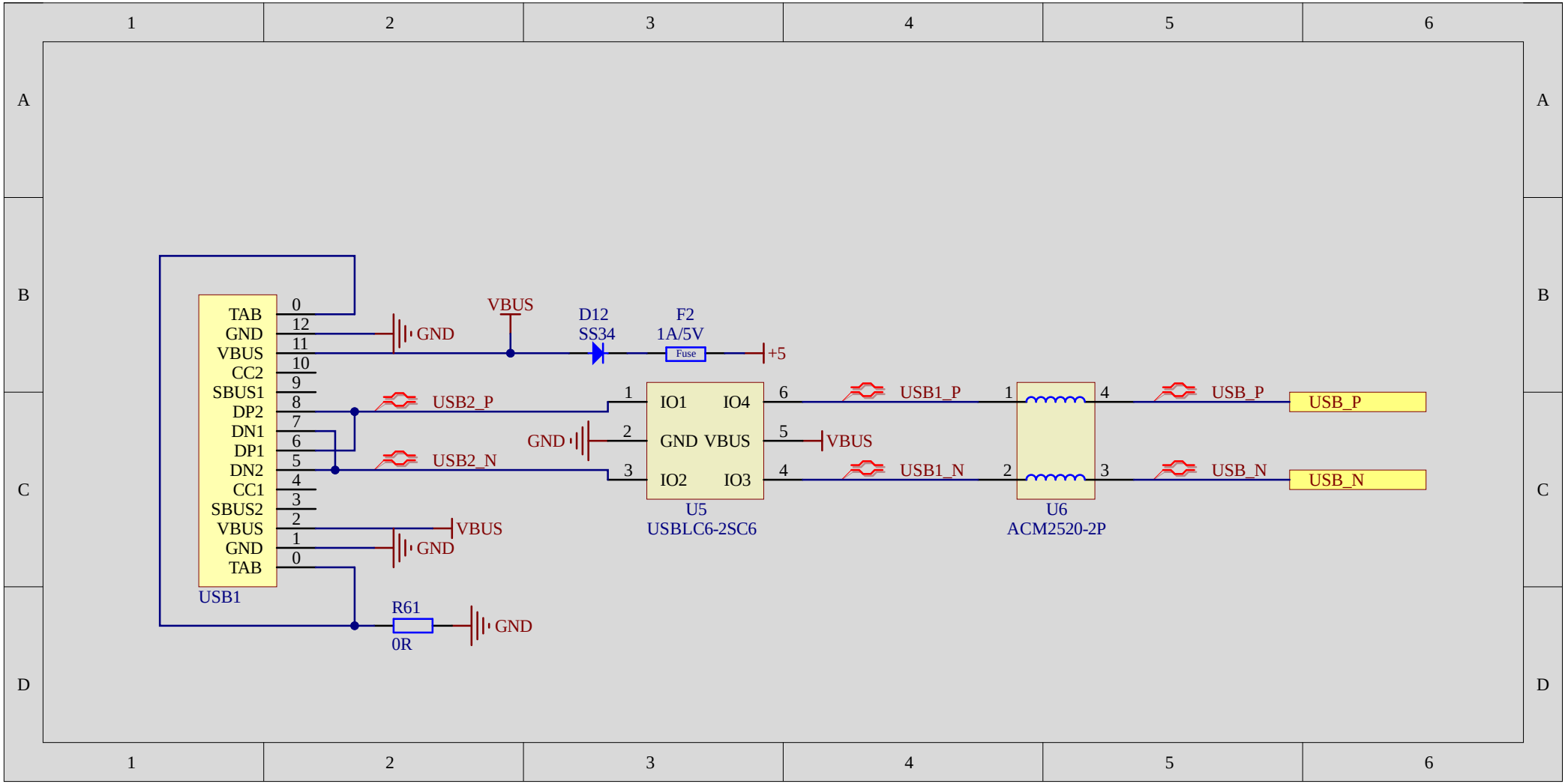

3V3
C2 4.7uF
C3 1uF
C4 0.1uF
C5 4.7uF
C6 1uF
C7 0.1uF
C97 4.7uF
C98 1uF
C99 0.1uF
C8 4.7uF
C9 1uF
C10 0.1uF
C11 4.7uF
C12 1uF
C13 0.1uF
C14 4.7uF
C15 1uF
C16 0.1uF
VBUS
C17 4.7uF
C18 1uF
C19 0.1uF
VREF+
C20 4.7uF
C21 1uF
C22 0.1uF
R6 10K
NRST
C1 0.1uF
BT1
C23 2.2uF
VCAP1
C24 2.2uF
VCAP2
LED1 RED
R1 10K
PE0
LED2 RED
R2 10K
PE15
LED3 RED
R3 10K
PD15
BT2
BOOT
R78 10K
C25 0.1uF
C26 7pF
XT3 32.768Khz
OSC32_P
C27 7pF
OSC32_N
C28 10pF
XT4 16Mhz
OSC_P
C29 10pF
OSC_N
CN7
SWCLK
SWDIO
CN4
MCU1
ETH
ETH_REF_CLK
ETH_MDIO
ETH_TX_EN
ETH_CRS_DV
ETH_TXD0
ETH_TXD1
ETH_MDC
ETH_RXD0
ETH_RXD1
SPI_NSS
SPI_SCK
SPI_MISO
SPI_MOSI
USB_P
USB_N
USART6_DE
USART6_RX
USART6_TX
USART10_DE
USART10_TX
USART10_RX
USART3_DE
USART3_RX
USART3_TX
A7680C_RX
A7680C_TX
A7680C_PWR
A7680C_RST
ST67_RDY
ST67_EN
ST67_BOOT
ST67_Wakeup
ST67_32K
ST67_RX
ST67_TX
ST67_NCS
ST67_SCK
ST67_MISO
ST67_MOSI
OCTOSPI
OCTOSPI_IO0
OCTOSPI_IO1
OCTOSPI_IO2
OCTOSPI_IO3
OCTOSPI_IO4
OCTOSPI_IO5
OCTOSPI_IO6
OCTOSPI_IO7
OCTOSPI_CLK
OCTOSPI_NCS
OCTOSPI_DQS
OCTOSPI_IO[0..7]
R77 47R
VREF+
GND

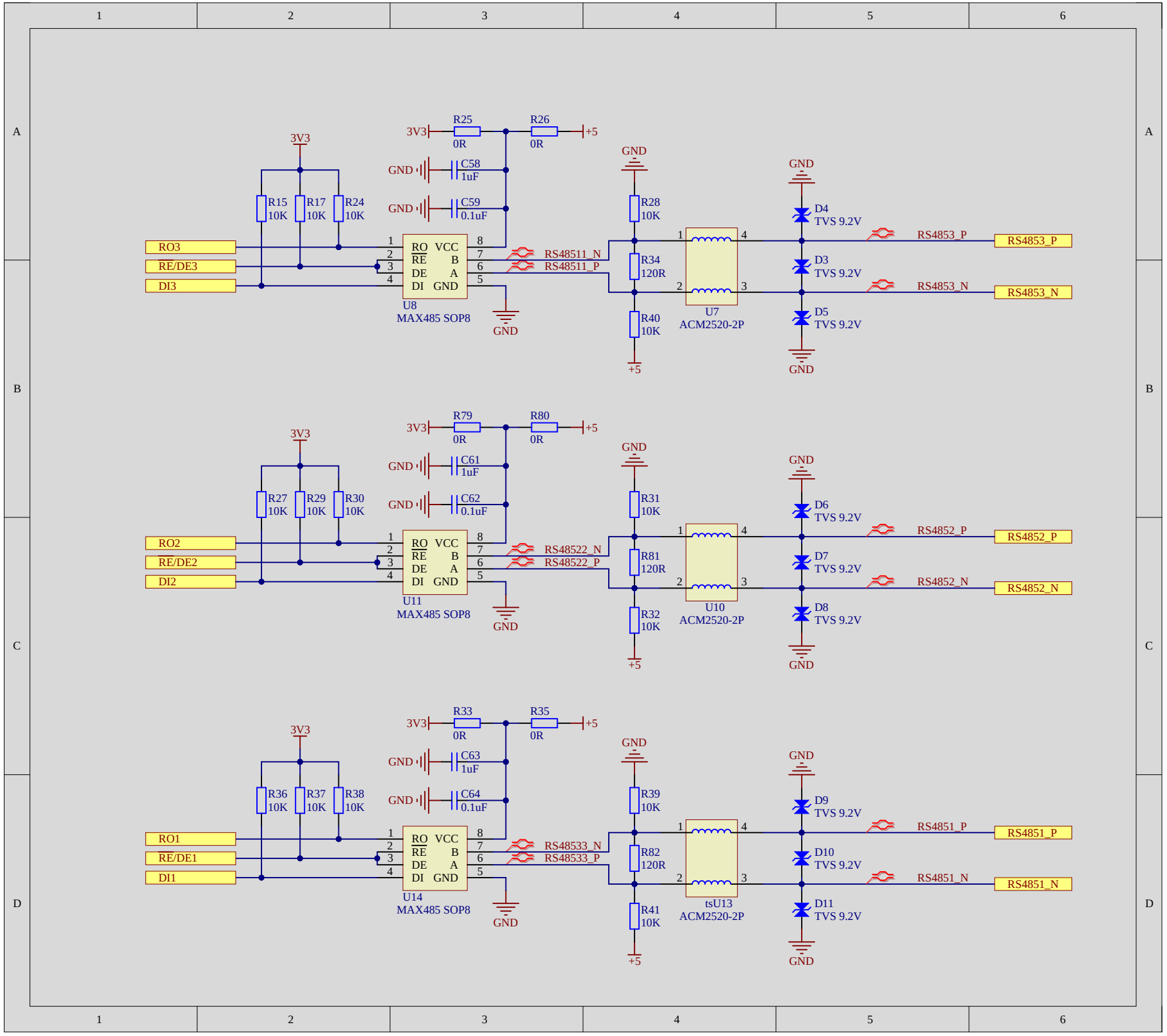


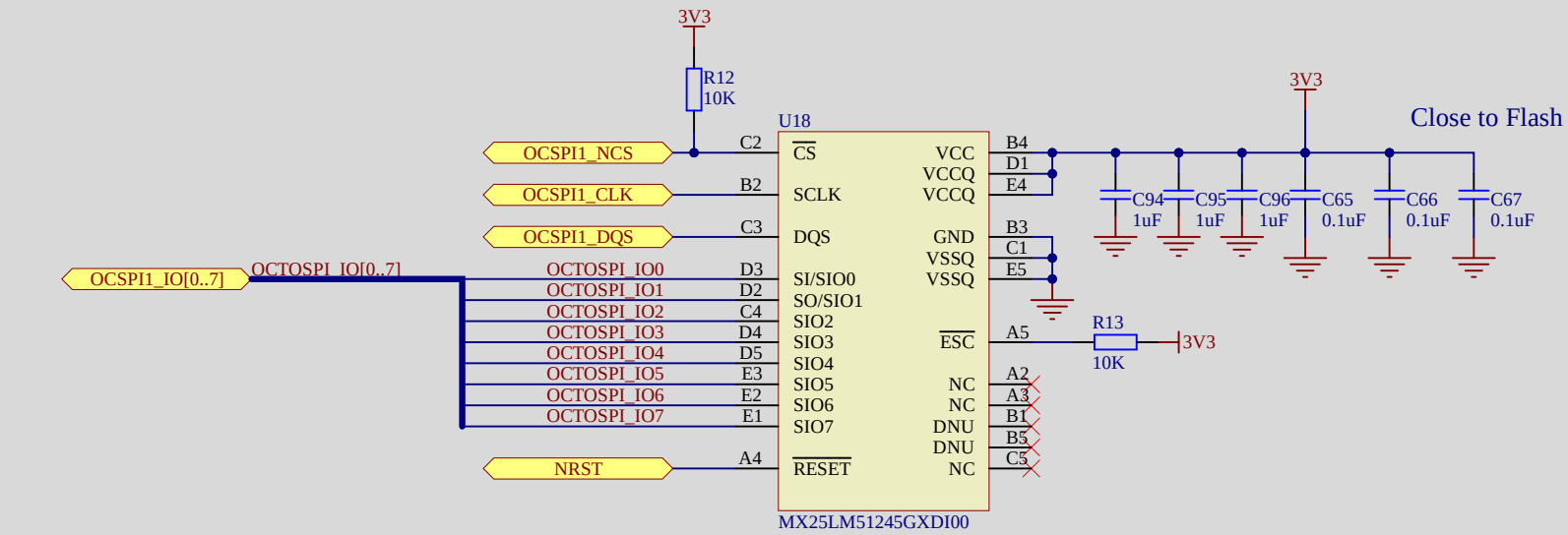
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Size A	Number	Revision
Date: 7/06/2026		Sheet of
File: D:\OneDrive - MB376106-Cty CP Havicom\Havicom\..\Wifi.SchDoc		Drawn By:


+24
FUSE1
Fuse
1A/24V
D1
SS34
D2
TVS 24V
GND
C45
10uF/35V
C37
4.7uF
C38
0.1pF
U3
TPS54302 SOT23-6
VIN
SW
EN
BOOT
GND
FB
C36
0.1uF
L1
10uH
C39
22uF
C40
22uF
R14
49.9R
R16
100K
R18
22.1K
C52
75pF
C51
100uF/10V
R59
10K
LD3
3V3
3V3
C54
10uF/35V
C43
4.7uF
C44
0.1pF
U4
TPS54302 SOT23-6
C42
0.1uF
L2
10uH
C46
22uF
C47
22uF
R19
49.9R
R21
100K
R23
13.3K
C57
75pF
C60
100uF/16V
R60
10K
LD4
+5V
+5

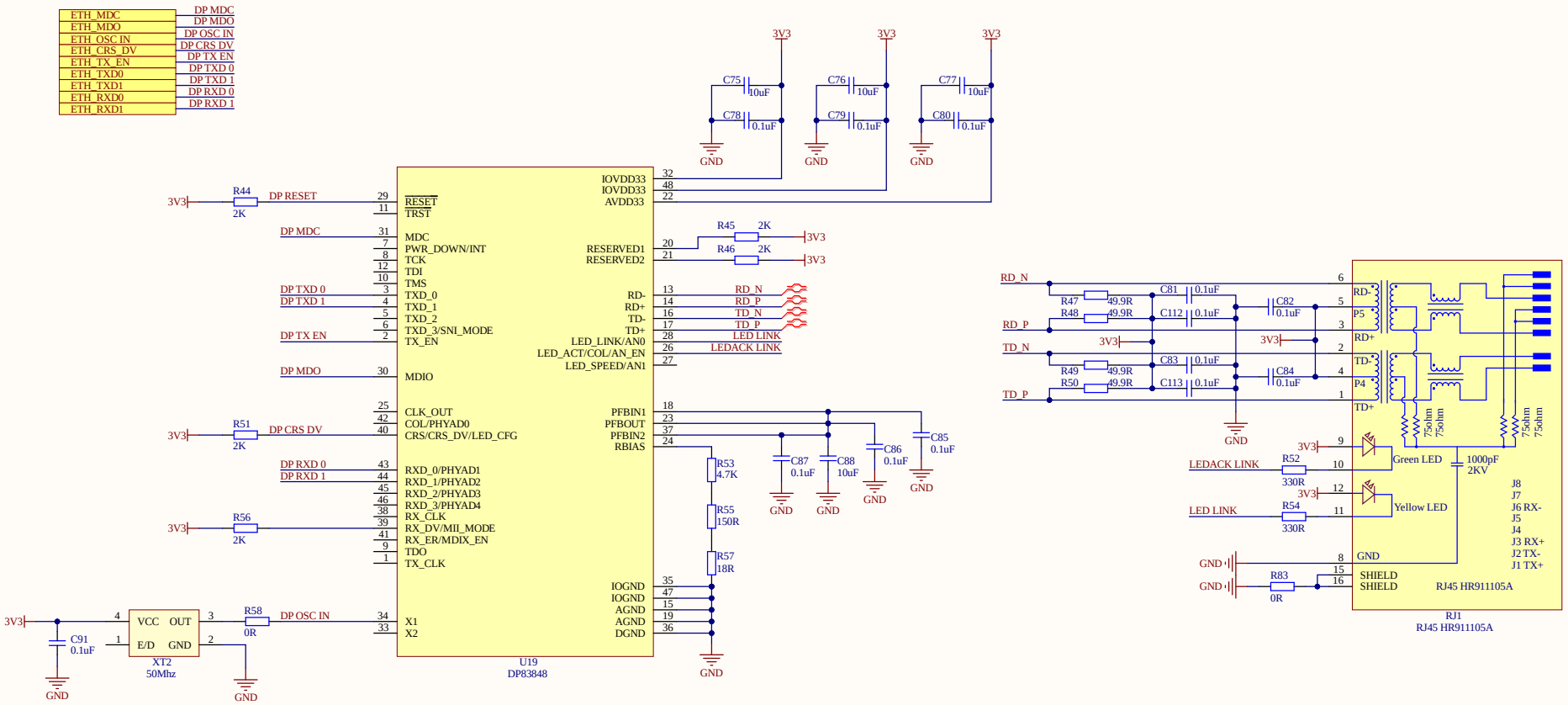

1
2
3
4
5
6
A
B
C
D
1
2
GND
+24
CN1
1
2
RS4851_P
RS4851_N
CN2
1
2
RS4852_P
RS4852_N
CN3
1
2
RS4853_P
RS4853_N
CN5

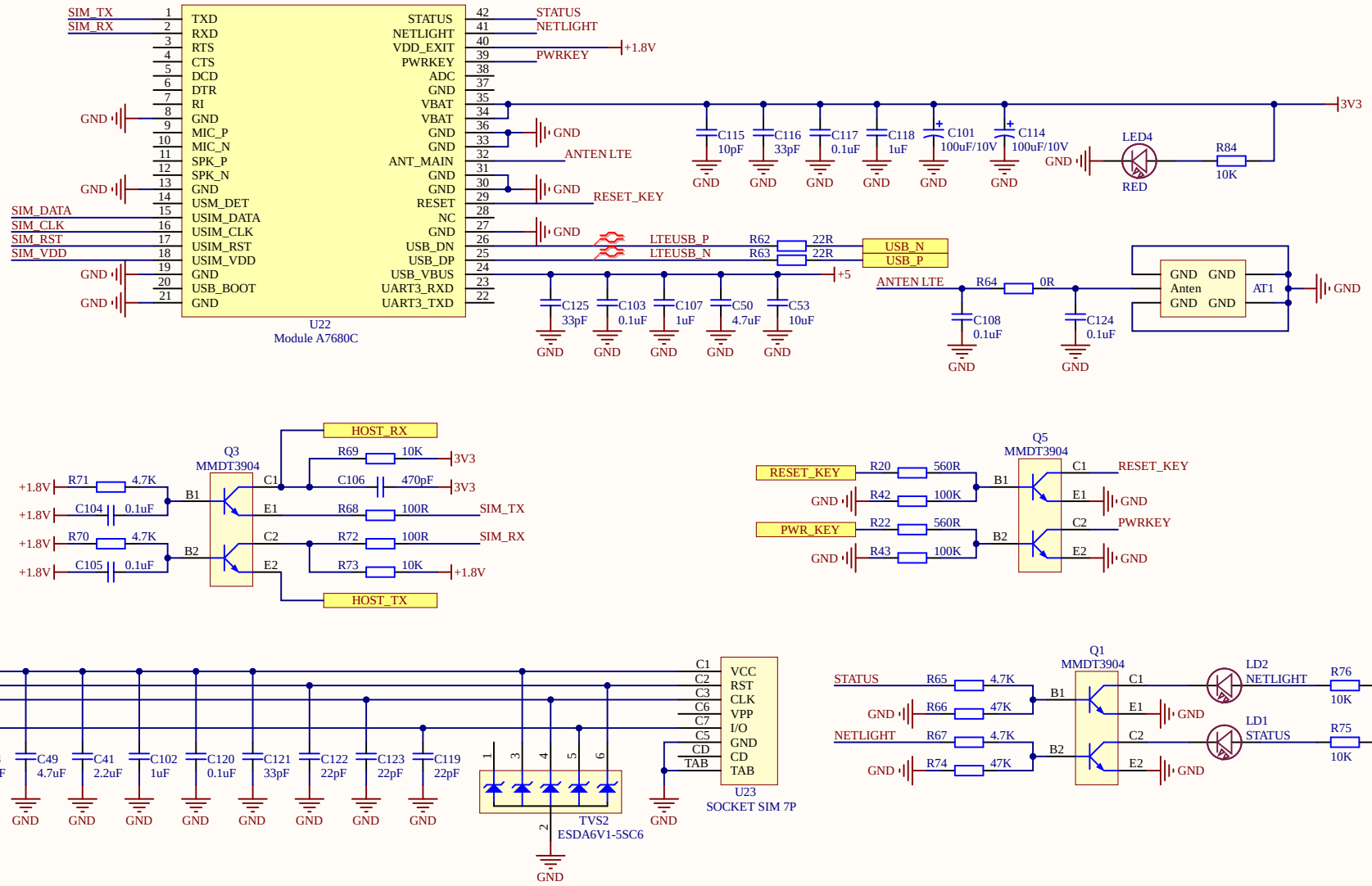

1
2
3
4
5
6
A
B
C
D
TAB
GND
VBUS
CC2
SBUS1
DP2
DN1
DP1
DN2
CC1
SBUS2
VBUS
GND
TAB
0
12
11
10
9
8
7
6
5
4
3
2
1
0
USB1
GND
VBUS
D12
SS34
F2
1A/5V
Fuse
+5
USB2_P
USB2_N
GND
VBUS
GND
R61
0R
GND
1
2
3
IO1
GND
IO2
IO4
VBUS
IO3
6
5
4
U5
USBLC6-2SC6
USB1_P
USB1_N
1
2
3
4
U6
ACM2520-2P
USB_P
USB_N
USB_P
USB_N


3V3
R25
0R
R26
0R
+5
GND
C58
1uF
C59
0.1uF
R15
10K
R17
10K
R24
10K
RO3
RE/DE3
DI3
RO
RE
DE
DI
VCC
B
A
GND
U8
MAX485 SOP8
RS48511_N
RS48511_P
R28
10K
R34
120R
R40
10K
U7
ACM2520-2P
D4
TVS 9.2V
D3
TVS 9.2V
D5
TVS 9.2V
RS4853_P
RS4853_N
R79
0R
R80
0R
C61
1uF
C62
0.1uF
R27
10K
R29
10K
R30
10K
RO2
RE/DE2
DI2
U11
MAX485 SOP8
RS48522_N
RS48522_P
R31
10K
R81
120R
R32
10K
U10
ACM2520-2P
D6
TVS 9.2V
D7
TVS 9.2V
D8
TVS 9.2V
RS4852_P
RS4852_N
R33
0R
R35
0R
C63
1uF
C64
0.1uF
R36
10K
R37
10K
R38
10K
RO1
RE/DE1
DI1
U14
MAX485 SOP8
RS48533_N
RS48533_P
R39
10K
R82
120R
R41
10K
tsU13
ACM2520-2P
D9
TVS 9.2V
D10
TVS 9.2V
D11
TVS 9.2V
RS4851_P
RS4851_N



Title		
Size: A	Number	Revision
Date: 7/06/2026		Sheet of
File: D:\OneDrive - MB376106-Cty CP Havicom\Havicom\..\Ethernet Wifi Gateway - SPI.SchDoc		Drawn By:


ETH_MDC
ETH_MDO
ETH_OSC IN
ETH_CRS_DV
ETH_TX_EN
ETH_TXD0
ETH_TXD1
ETH_RXD0
ETH_RXD1
DP MDC
DP MDO
DP OSC IN
DP CRS DV
DP TX EN
DP TXD 0
DP TXD 1
DP RXD 0
DP RXD 1
3V3
C75 10uF
C78 0.1uF
C76 10uF
C79 0.1uF
C77 10uF
C80 0.1uF
GND
R44 2K
DP RESET
RESET
TRST
MDC
PWR_DOWN/INT
TCK
TDI
TMS
TXD_0
TXD_1
TXD_2
TXD_3/SNI_MODE
TX_EN
MDIO
R51 2K
CRS/CRS_DV/LED_CFG
R56 2K
RXD_0/PHYAD1
RXD_1/PHYAD2
RXD_2/PHYAD3
RXD_3/PHYAD4
RX_CLK
RX_DV/MII_MODE
RX_ER/MDIX_EN
TDO
TX_CLK
CLK_OUT
COL/PHYAD0
R58 0R
X1
X2
XT2
50Mhz
C91 0.1uF
U19
DP83848
IOVDD33
IOVDD33
AVDD33
R45 2K
R46 2K
RESERVED1
RESERVED2
RD-
RD+
TD-
TD+
LED_LINK/AN0
LED_ACT/COL/AN_EN
LED_SPEED/AN1
RD_N
RD_P
TD_N
TD_P
LED LINK
LEDACK LINK
PFBIN1
PFBOUT
PFBIN2
RBIAS
R53 4.7K
R55 150R
R57 18R
C87 0.1uF
C88 10uF
C86 0.1uF
C85 0.1uF
IOGND
IOGND
AGND
AGND
DGND
C81 0.1uF
C112 0.1uF
C83 0.1uF
C113 0.1uF
R47 49.9R
R48 49.9R
R49 49.9R
R50 49.9R
C82 0.1uF
C84 0.1uF
750ohm
R52 330R
R54 330R
Green LED
Yellow LED
1000pF 2KV
R83 0R
SHIELD
RJ45 HR911105A
J8
J7
J6 RX-
J5
J4
J3 RX+
J2 TX-
J1 TX+
RJ1
RJ45 HR911105A


SIM_TX
SIM_RX
SIM_DATA
SIM_CLK
SIM_RST
SIM_VDD
TXD
RXD
RTS
CTS
DCD
DTR
RI
GND
MIC_P
MIC_N
SPK_P
SPK_N
USM_DET
USIM_DATA
USIM_CLK
USIM_RST
USIM_VDD
USB_BOOT
STATUS
NETLIGHT
VDD_EXIT
PWRKEY
ADC
VBAT
ANT_MAIN
RESET
NC
USB_DN
USB_DP
USB_VBUS
UART3_RXD
UART3_TXD
U22
Module A7680C
ANTEN LTE
RESET_KEY
LTEUSB_P
LTEUSB_N
R62 22R
R63 22R
USB_N
USB_P
+1.8V
+5
3V3
C115 10pF
C116 33pF
C117 0.1uF
C118 1uF
C101 100uF/10V
C114 100uF/10V
LED4
RED
R84 10K
C125 33pF
C103 0.1uF
C107 1uF
C50 4.7uF
C53 10uF
R64 0R
C108 0.1uF
C124 0.1uF
GND Anten GND GND
AT1
HOST_RX
HOST_TX
Q3
MMDT3904
R71 4.7K
R70 4.7K
C104 0.1uF
C105 0.1uF
R69 10K
C106 470pF
R68 100R
R72 100R
R73 10K
SIM_TX
SIM_RX
Q5
MMDT3904
R20 560R
R42 100K
R22 560R
R43 100K
C48 10uF
C49 4.7uF
C41 2.2uF
C102 1uF
C120 0.1uF
C121 33pF
C122 22pF
C123 22pF
C119 22pF
TVS2
ESDA6V1-5SC6
U23
SOCKET SIM 7P
Q1
MMDT3904
STATUS
NETLIGHT
R65 4.7K
R66 47K
R67 4.7K
R74 47K
LD2
NETLIGHT
LD1
STATUS
R76 10K
R75 10K

Wifi RS485
Version 1.0
JTMS
JTCK
JTDI
BOOT
GND
+3V3
SWDO
SWCL
RJ1
USB1
FUSE1
MCU1
BT1
BT2
AT1
AT2
LED1
LED2
LED3
LED4
TVS2
U23

Comment	Description	Designator	Footprint	LibRef	Quantity
Anten SMA-KWE	Anten SMA-KWE	AT1, AT2	ANTEN SMA-KWE-01	Anten SMA-KWE	2
BUTTON 2P	BUTTON 2P	BT1, BT2	BUTTON SMD-RIGHT 2P 3x6x3.5	BUTTON 2P	2
0.1uF	CAP CERAMIC, 100nF-0603, 100nF, 10uF-0603	C1, C4, C7, C10, C13, C16, C19, C22, C25, C32, C36, C42, C55, C59, C62, C64, C65, C66, C67, C68, C72, C78, C79, C80, C81, C82, C83, C84, C85, C86, C87, C90, C91, C99, C103, C104, C105, C108, C112, C113, C117, C120, C124	CAP 0603 PAD MEDIUM	CAP CERAMIC	43
4.7uF	CAP CERAMIC, 10uF-0603	C2, C5, C8, C11, C14, C17, C20, C34, C37, C43, C49, C50, C70, C74, C92, C97	CAP 0603 PAD MEDIUM	CAP CERAMIC	16
1uF	CAP CERAMIC, 10uF-0603, 100nF-0603	C3, C6, C9, C12, C15, C18, C21, C35, C58, C61, C63, C69, C71, C89, C94, C95, C96, C98, C102, C107, C118	CAP 0603 PAD MEDIUM	CAP CERAMIC	21
2.2uF	CAP CERAMIC, 10uF-0603	C23, C24, C41	CAP 0603 PAD MEDIUM	CAP CERAMIC	3
7pF	CAP CERAMIC	C26, C27	CAP 0603 PAD MEDIUM	CAP CERAMIC	2
10pF	CAP CERAMIC, 10uF-0603	C28, C29, C115	CAP 0603 PAD MEDIUM	CAP CERAMIC	3
5.1pF	CAP CERAMIC	C30, C31	CAP 0603 PAD MEDIUM	CAP CERAMIC	2
10uF	CAP CERAMIC, 10uF-0603	C33, C48, C53, C56, C73, C75, C76, C77, C88, C93	CAP 0603 PAD MEDIUM	CAP CERAMIC	10
0.1pF	CAP CERAMIC	C38, C44	CAP 0603 PAD MEDIUM	CAP CERAMIC	2
22uF	CAP CERAMIC	C39, C40, C46, C47	CAP 0603 PAD MEDIUM	CAP CERAMIC	4
10uF/35V	10uF/35V-6032	C45, C54	CAP TANTALUM SMD 6032	CAP TANTALUM	2
100uF/10V	100uF/16V-6032, CAP TANTALUM	C51, C101, C114	CAP TANTALUM SMD 3528	CAP TANTALUM	3
75pF	CAP CERAMIC	C52, C57	CAP 0603 PAD MEDIUM	CAP CERAMIC	2
100uF/16V	100uF/16V-6032	C60	CAP TANTALUM SMD 3528	CAP TANTALUM	1
470pF	10uF-0603	C106	CAP 0603 PAD MEDIUM	CAP CERAMIC	1
33pF	100nF-0603, 10uF-0603	C116, C121, C125	CAP 0603 PAD MEDIUM	CAP CERAMIC	3
22pF	100nF-0603	C119, C122, C123	CAP 0603 PAD MEDIUM	CAP CERAMIC	3
Connect 2P	Connect 2P	CN1, CN2, CN3, CN5	Connector 2EDG 2P 5.08 DIP RIGHT	Connect 2P	4
Connect 30P-X2	Connect 30P-X2	CN4	JUMP DIP 2.54 15P-X2	Connect 15P-X2	1
Connect 6P	Connect 6P	CN6	JUMP DIP 2.54 6P-X1	Connect 6P	1
Connect 2P	Connect 2P	CN7	JUMP DIP 2.54 2P-X1	Connect 2P	1
SS34	DIODE SCHOTTKY 2P	D1, D12	DIODE SCHOTTKY SOD123	DIODE SCHOTTKY 2P	2
TVS 24V	DIODE TVS SMD 2P	D2	DIODE TVS SMA	DIODE TVS SMD 2P	1
TVS 9.2V	DIODE TVS SMD 2P	D3, D4, D5, D6, D7, D8, D9, D10, D11	DIODE TVS SOD323	DIODE TVS SMD 2P	9
1A/5V	FUSE	F2	FUSE 1206 PAD SMALL	FUSE	1
1A/24V	FUSE	FUSE1	FUSE 1206 PAD SMALL	FUSE	1
10uH	INDUCTOR SMD	L1, L2	INDUCTOR SMD 4.3x4.3	INDUCTOR SMD	2
STATUS	LED 2P	LD1	LED 0603 PAD MEDIUM	LED 2P	1
NETLIGHT	LED 2P	LD2	LED 0603 PAD MEDIUM	LED 2P	1
3V3	LED 2P	LD3	LED 0603 PAD MEDIUM	LED 2P	1
+5V	LED 2P	LD4	LED 0603 PAD MEDIUM	LED 2P	1
RED	LED 2P	LED1, LED2, LED3	LED DIP 3MM	LED 2P	3
RED	LED 2P	LED4	LED 0603 PAD MEDIUM	LED 2P	1
STM32H563VGT6 LQFP100	STM32H563VGT6 LQFP100	MCU1	STM32H563VGT6 LQFP100	STM32H563VGT6 LQFP100	1
MMDT3904	Transistor NPN	Q1, Q3, Q5	MMDT3904 SOT23-6	Transistor 2CH NPN	3
10K	2K-0603, 100K-0603, RES	R1, R2, R3, R6, R12, R13, R15, R17, R24, R27, R28, R29, R30, R31, R32, R36, R37, R38, R39, R40, R41, R59, R60, R69, R73, R75, R76, R78, R84, R85	RES 0603 PAD MEDIUM	RES SMD, RES	30
0R	RES, 0R-0603, 2K-0603, 330R-0603	R4, R5, R7, R8, R10, R11, R25, R26, R33, R35, R58, R61, R64, R79, R80, R83	RES 0603 PAD MEDIUM	RES, RES SMD	16
33K	RES	R9	RES 0603 PAD MEDIUM	RES	1
49.9R	49.9R-0603	R14, R19, R47, R48, R49, R50	RES 0603 PAD MEDIUM	RES SMD	6
100K	100K-0603, 2K-0603	R16, R21, R42, R43	RES 0603 PAD MEDIUM	RES SMD	4
22.1K	13.3K-0603	R18	RES 0603 PAD MEDIUM	RES SMD	1
560R	2K-0603	R20, R22	RES 0603 PAD MEDIUM	RES SMD	2
13.3K	13.3K-0603	R23	RES 0603 PAD MEDIUM	RES SMD	1
120R	RES	R34, RB1, RB2	RES 0603 PAD MEDIUM	RES	3
2K	2K-0603	R44, R45, R46, R51, R56	RES 0603 PAD MEDIUM	RES SMD	5
330R	330R-0603	R52, R54	RES 0603 PAD MEDIUM	RES SMD	2
4.7K	4.7K-0603, 2K-0603	R53, R65, R67, R70, R71	RES 0603 PAD MEDIUM	RES SMD	5
150R	150R-0603	R55	RES 0603 PAD MEDIUM	RES SMD	1
18R	18R-0603	R57	RES 0603 PAD MEDIUM	RES SMD	1
22R	2K-0603	R62, R63	RES 0603 PAD MEDIUM	RES SMD	2
47K	2K-0603	R66, R74	RES 0603 PAD MEDIUM	RES SMD	2
100R	2K-0603	R68, R72	RES 0603 PAD MEDIUM	RES SMD	2
47R	2K-0603	R77	RES 0603 PAD MEDIUM	RES SMD	1
RJ45 HR911105A	RJ45 HR911105A	RJ1	RJ45 HR911105A	RJ45 HR911105A	1
ACM2520-2P	Filter Roll ACM2520-2P	tsU13, U6, U7, U10	Filter Roll ACM2520-2P	Filter Roll ACM2520-2P	4
ESDA6V1-5SC6	ESDA6V1-5SC6 SOT23-6	TVS2	ESDA6V1-5SC6 SOT23-6	ESDA6V1-5SC6 SOT23-6	1
ST67W611M1A6UTR	Low-power Wi-Fi 6 & Bluetooth LE combo coprocessor module	U2	U_QFN32L_ST67W611M1_STM	ST67W611M1A6UTR	1
TPS54302 SOT23-6	TPS54302 SOT23-6	U3, U4	TPS54302 SOT23-6	TPS54302 SOT23-6	2
USBLC6-2SC6	DIODE USBLC6-2SC6 SOT23-6	U5	DIODE USBLC6-2SC6 SOT23-6	DIODE USBLC6-2SC6 SOT23-6	1
MAX485 SOP8	MAX485 SOP8	U8, U11, U14	MAX485 SOP8	MAX485 SOP8	3
W25Q128	W25Q128	U16	W25Q128 SOIC8	W25Q128	1
MX25LM51245GXDI00		U18	MX25LM51245GXDI00	MX25LM51245GXDI00	1
DP83848	DP83848	U19	DP83848 LQFP48	DP83848	1
Module A7680C	Module A7680C	U22	Module A7680C	Module A7680C	1
SOCKET SIM 7P	SOCKET SIM 7P	U23	SOCKET NANO SIM 7P	SOCKET SIM 7P	1
SOCKET USB TYPE-C SMD 16PIN RIGHT	SOCKET USB TYPE-C SMD 16PIN RIGHT	USB1	SOCKET USB TYPE-C SMD 16PIN RIGHT	SOCKET USB TYPE-C SMD 16PIN RIGHT	1
32.768Khz	CRYATAL 2P	XT1, XT3	CRYSTAL 2P FC-135	CRYATAL 2P	2
50Mhz	CRYSTAL OSCILLATOR 4P	XT2	CRYSTAL OSCILLATOR 4P 5032	CRYSTAL OSCILLATOR 4P	1
16Mhz	CRYATAL 2P	XT4	CRYSTAL 2P DIP 495	CRYATAL 2P	1